Positive Square Energy of Every Tetracyclic Cactus

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Abstract

Let G be a connected tetracyclic cactus of order n. We prove $s^+(G) > n$. The sharp block-additive doubly nonnegative bound reduces the problem to cycle multisets $\{3, 3, 3, q\}$ and $\{3, 3, 5, 5\}$. We settle every bridge-separated incidence by adaptive induced packetization. Two apparent obstructions disappear only after one allows a hostile cycle to be sacrificed into path fragments: a shared $C_3 - C_5 - C_3$ cluster plus a remote pentagon, and a shared three-triangle cluster plus a remote 1 (mod 4) cycle. Arbitrary connector trees and arbitrary trees attached at arbitrary vertices are allowed. Fully shared residuals are closed by deleting a private cyclic vertex or by partitioning a hostile central cycle into three triangular unicyclic territories.

1 Preliminaries

All graphs are finite, simple, and undirected. If the adjacency eigenvalues are $\lambda_1, \ldots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad \sigma(G) = s^+(G) - |V(G)|.$$

We use induced-subgraph superadditivity [1]: for every vertex partition $V(G) = V_1 \dot\cup \cdots \dot\cup V_k$,

$$s^+(G) \geq \sum_{j=1}^{k} s^+(G[V_j]). \tag{1}$$

Join two cyclic blocks in the *shared-cut graph* when they share a cut vertex, and call its connected components shared-cut clusters. In the block-cut tree, contract the incidence subtree belonging to each shared-cut cluster to one marked node; every cluster, including every singleton cyclic block, is marked. Take the minimal subtree spanning these marked nodes and suppress only unmarked degree-two nodes. No marked node is ever suppressed. The resulting tree is the reduced cluster tree. Consequently the interior of every reduced edge contains only bridge blocks and unmarked cut nodes, never an unassigned cyclic block.

Lemma 1.1 (Connector territories)**.** *Let $S_1, \ldots, S_k$ be pairwise vertex-disjoint connected subtrees of the reduced cluster tree such that every marked cluster node belongs to exactly one S_i. Then there is a vertex partition into connected induced territories such that the territory corresponding to S_i retains exactly the cyclic blocks in the clusters marked in S_i.*

Proof. Every reduced edge represents a path whose block nodes are all bridge blocks in the block-cut tree; this follows from the preceding convention that every shared-cut cluster, including a singleton cycle, is a nonsuppressed marked node. Assign the internal vertices of each selected connector path to its selected subtree. Between two selected subtrees, cut one actual bridge on their joining path and assign the two remnants to their respective sides. At an unmarked Steiner node, assign the node and its initial path segments to one selected subtree and cut one bridge on every other incident segment. Every component outside the minimal cyclic hull meets it at a unique vertex, since two attachments would create a cycle in the block-cut tree; assign that entire hanging tree to the territory containing its attachment. The resulting sets are connected. Their induced subgraphs retain precisely the assigned cycles. □

We will also split a cyclic block by assigning a proper path of it to each of two territories. Crossing edges are harmless in (1).

We quote the following previously proved packet estimates [3, 4]. Each permits arbitrary tree attachments; the mixed bound permits arbitrary incidence and connectors, while (5) and (6) have their displayed shared-cut hypotheses. Write $T = C_3$, $P = C_5$, and, for $q \equiv 1 \pmod 4$,

$$\delta_q = \sec(\pi/q) - 1 < 1, \qquad \delta_5 = \sqrt{5} - 2.$$

Then

$$\sigma(T) > 0, \qquad \sigma(C_q) \geq -\delta_q, \tag{2}$$

$$\sigma(TT) > 1, \qquad \sigma(TC_q) > 1 - \delta_q, \tag{3}$$

$$\sigma(H) \geq 0 \qquad \text{for every bicyclic or tricyclic cactus } H, \tag{4}$$

$$\sigma(TTC_q) > 2 - \delta_q \qquad \text{if the two triangles share a cut vertex,} \tag{5}$$

$$\sigma(TPP) > 6 - 2\sqrt{5} \qquad \text{if the three cycles form one shared-cut cluster.} \tag{6}$$

If every cycle is 3 (mod 4) and the cycle-packing number is at most two, the Sachs half-plane theorem [6] gives $D = s^+ - s^- > 0$. Thus a connected r-cyclic such cactus has

$$\sigma(G) > r - 1. \tag{7}$$

These results are proved in the companion manuscripts on all bicyclic and all tricyclic cacti.

2 The exact residual reduction

For a cactus with bridge-block count b and cycle lengths $\ell_1, \ldots, \ell_4$, block counting gives

$$b + \sum_{j=1}^{4} \ell_j = n + 3.$$

The sharp cactus DNN constant [5, 7] implies

$$s^+(G) - n \geq 3 - \sum_{j=1}^{4} \epsilon_{\ell_j}, \tag{8}$$

where $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ. Moreover

$$\epsilon_3 = 1, \qquad \epsilon_5 = 5 - 2\sqrt{5}, \qquad \epsilon_5 + \epsilon_7 < 1,$$

and ϵ_ℓ strictly decreases through odd lengths. Hence (8) proves the theorem unless the cycle multiset is

$$\{3, 3, 3, q\} \quad (q \geq 3), \qquad \text{or} \qquad \{3, 3, 5, 5\}. \tag{9}$$

3 Three triangles and one further cycle

Proposition 3.1. *Let the cyclic blocks of G have lengths $3, 3, 3, q$, where $q \geq 3$. If the shared-cut graph is disconnected, then $s^+(G) > n$.*

Proof. If q is even, take the shared-cut clusters as induced territories using Lemma 1.1. The territory containing the even cycle is at most tricyclic and has nonnegative surplus by (4) (and is bipartite with zero surplus if it contains no triangle). Since the shared-cut graph is disconnected, another territory contains one, two, or three triangles. Its cycle-packing number is at most two, so its surplus is strictly positive by (7). Every remaining territory has nonnegative surplus. Equation (1) proves the claim.

If $q \equiv 3 \pmod 4$, partition along the bridge connectors between shared-cut clusters. Every territory contains only favorable cycles and has cycle packing at most two within each nontrivial cluster; singleton territories are favorable unicyclic cacti. Equations (7) and (1) give the result.

Suppose $q \equiv 1 \pmod 4$, and denote its block by Q. Up to permutation of the triangles, the shared-cut component partitions are

$$T|T|T|Q, \quad TT|T|Q, \quad TQ|T|T, \quad TTT|Q, \quad TT|TQ, \quad TTQ|T.$$

We treat them in turn.

For $T|T|T|Q$, choose a triangle whose marked node has minimum distance from Q in the reduced cluster tree. The unique Q–T path then contains no other marked node. Use this path as one connected subtree and the other two triangle nodes as singleton subtrees; Lemma 1.1 realizes the territories even when the path passes through an unmarked Steiner node. By (2)–(3), the total surplus is strictly greater than $1 - \delta_q > 0$. The same direct accounting settles $TQ|T|T$. The partition $TT|TQ$ has surplus greater than $1 + (1 - \delta_q) > 0$.

For $TT|T|Q$, call the three marked nodes $A = TT$, $B = T$, and $C = Q$. If the B–C path avoids A, Lemma 1.1 gives a TQ packet and a separate TT packet. Otherwise A lies on that path; the connected A–C subtree gives a TTQ packet separated from B. The resulting TTQ territory retains the shared pair of triangles, so (5), together with the remaining triangular packet, gives total surplus greater than $2 - \delta_q > 0$.

For $TTQ|T$, the tricyclic territory has nonnegative surplus by (4), and the singleton triangle has strictly positive surplus.

It remains to treat $TTT|Q$. Let $w \in V(Q)$ be the vertex through which the block-tree route from Q to the triangular cluster leaves Q. Choose $v \in V(Q) \setminus \{w\}$. Let F consist of v together with every component of $G - E(Q)$ attached to Q only at v, and put all other vertices into H. The graph $Q - v$ is a path containing w, so H is connected; its only cycles are the three triangles in one shared-cut cluster. Thus its cycle-packing number is at most two and

$$s^+(H) > |H| + 2$$

by (7). Since F is a tree, $s^+(F) = |F| - 1$. Therefore

$$s^+(G) \geq s^+(H) + s^+(F) > |H| + 2 + |F| - 1 = n + 1.$$

All component partitions have now been covered. □

4 Two triangles and two pentagons

The only delicate packetization is recorded separately.

Lemma 4.1 (Middle-pentagon sacrifice). *Suppose one shared-cut cluster consists of a pentagon P and two disjoint triangles T_1, T_2 meeting P at distinct vertices x_1, x_2. Suppose a second pentagon Q is bridge-separated from this cluster. Then the vertices can be partitioned into connected induced territories which are either*

1. *a TT bicyclic cactus and a pentagonal unicyclic cactus, or*
2. *a triangular unicyclic cactus and a TP bicyclic cactus.*

In either case $s^+(G) > n$.

Proof. Let z be the first vertex of the shared cluster met by the unique block-tree route from Q. Include Q, its entire connector, and all branches hanging from that connector on the Q side.

If $z \in V(P) \setminus \{x_1, x_2\}$, put z with Q and put $P - z$ together with both triangles on the other side. The latter is connected because $P - z$ is a path containing x_1, x_2. The two induced territories have cycle multisets $\{Q\}$ and $\{T_1, T_2\}$, respectively. Every off-core branch rooted at z goes with the Q side. Hence their surplus sum is strictly greater than $-\delta_5 + 1 = 3 - \sqrt{5} > 0$.

Otherwise z lies on one triangle, say T_1; this includes $z = x_1$ (which also lies on P) and a connector entering through any branch rooted on T_1. Put all of T_1, the connector, and Q on one side, together with every off-core branch rooted at x_1. Put $P - x_1$ and T_2 on the other side. The first induced territory is a mixed TP bicyclic cactus; the second is triangular unicyclic because $P - x_1$ is a path containing x_2. Their surplus sum is strictly greater than $1 - \delta_5 > 0$. The case in which the route enters through T_2 is symmetric.

Every remaining bridge-tree component meets this retained skeleton at only one vertex; otherwise the block-cut graph would contain a cycle. Assign it wholly to the territory containing that attachment. Thus both territories remain connected and induced. In all cases the middle pentagon is deliberately split into proper path fragments, so it creates no uncounted cycle. □

Proposition 4.2. *Let the cyclic blocks of G have lengths $3, 3, 5, 5$. If the shared-cut graph is disconnected, then $s^+(G) > n$.*

Proof. Write again $T = C_3$ and $P = C_5$. Up to equal-cycle permutation, the possible shared-cut component partitions are

$$TTP|P, \quad TPP|T, \quad TT|PP, \quad TP|TP, \quad TT|P|P, \quad PP|T|T, \quad TP|T|P, \quad T|T|P|P.$$

For $TPP|T$, use (6) and the strict triangular surplus. For $TT|PP$ and $TP|TP$, the packet sums are respectively greater than 1 and $2(1 - \delta_5)$. For the three-cluster partitions, direct cluster territories give

$$\begin{aligned} TT|P|P: &\qquad \sigma(G) > 1 - 2\delta_5 = 5 - 2\sqrt{5} > 0, \\ PP|T|T: &\qquad \sigma(G) > 0, \\ TP|T|P: &\qquad \sigma(G) > 1 - 2\delta_5 > 0. \end{aligned}$$

For $T|T|P|P$, consider the reduced tree spanning the four cycle nodes. If a triangle is a leaf, cut its first bridge: the triangular packet has positive surplus and the remaining tricyclic cactus has nonnegative surplus. If neither triangle is a leaf, the only leaves are the two pentagons. A finite tree with exactly two leaves is a path, so the cycle order is $P - T - T - P$. Cut between the triangles to obtain two induced TP bicyclic territories, whose total surplus is greater than $2(1 - \delta_5) > 0$.

Finally consider $TTP|P$. If the two triangles in the triple cluster share a cut vertex, (5) gives total surplus greater than $2 - 2\delta_5 > 0$. Otherwise the shared-cut incidence tree is necessarily $T - P - T$, with the pentagon in the middle and distinct triangle cuts. This is exactly Lemma 4.1. The enumeration is exhaustive. □

5 Fully shared residuals

Let I denote the bipartite incidence tree whose nodes are the four cyclic blocks and the cut vertices belonging to at least two of them. If c is the number of cut nodes, then $|E(I)| = c + 3$ and

$$\sum_{x \text{ cut node}} (\deg_I(x) - 1) = 3. \tag{10}$$

Proposition 5.1 (Fully shared $\{3, 3, 5, 5\}$)**.** *If the four cyclic blocks have lengths* $3, 3, 5, 5$ *and form one shared-cut cluster, then* $s^+(G) > n$.

Proof. Suppose first that a triangle is a leaf cycle node of I, with unique shared cut x. Choose a private vertex v of that triangle, let F consist of v and every off-cycle tree branch rooted there, and put the remaining vertices in H. Then F is a tree, so $\sigma(F) = -1$. The broken triangle leaves an edge at x, while the other triangle and two pentagons remain connected in one shared-cut cluster. Thus (6) gives

$$\sigma(G) \geq \sigma(F) + \sigma(H) > -1 + 6 - 2\sqrt{5} = 5 - 2\sqrt{5} > 0.$$

Suppose neither triangle is a leaf. Every cut node has degree at least two, the two triangle nodes have degree at least two, and the pentagon nodes have degree at least one. Since $|E(I)| = c + 3$, degree counting gives $c + 3 \geq 2c$ and $c + 3 \geq 6$, hence $c = 3$. Equality holds throughout: every cut node has degree two, both triangles have degree two, and both pentagons are leaves. Therefore I is the alternating path

$$P_1 - x - T_1 - y - T_2 - z - P_2.$$

Choose a private vertex v of P_1 and define the tree F as above. The remainder H is connected, has cycles T_1, T_2, P_2, and its triangles share y. Equation (5) gives

$$\sigma(G) > -1 + (2 - \delta_5) = 1 - \delta_5 = 3 - \sqrt{5} > 0.$$

The two incidence alternatives are exhaustive. □

Lemma 5.2 (Three-petal partition)**.** *Suppose a cycle* $Q = C_q$ *carries three pairwise disjoint triangular blocks at three distinct vertices. Then the vertices can be partitioned into three connected induced triangular unicyclic cacti.*

Proof. List the three attachment vertices in cyclic order. Partition $V(Q)$ into three nonempty consecutive cyclic intervals, each containing exactly one attachment vertex; equivalently, choose the three boundary edges between successive intervals. If two attachments are adjacent, their common edge may be a boundary edge, and for $q = 3$ all intervals are singletons. Every interval induces a proper path, never the whole cycle. Add to each interval its attached triangle. Every component outside the four-cycle core is a tree with a unique core attachment, since all four cyclic blocks have already been listed; assign it wholly to the part containing that attachment. Each resulting induced graph is connected and its unique cycle is its triangle. □

Proposition 5.3 (Fully shared $\{3, 3, 3, q\}$)**.** *If the four cyclic blocks have lengths* $3, 3, 3, q$, $q \geq 3$, *and form one shared-cut cluster, then* $s^+(G) > n$.

Proof. When $q = 3$, all four blocks are interchangeable; label them so that any chosen intersecting pair is among the three designated triangles. Suppose first that two designated triangles share a cut. Let k be the number of shared cut vertices on Q. The k cuts contribute at least k to (10);

the intersecting pair contributes one further unit, whether its cut lies off Q or is a degree-at-least-three cut on Q. Hence $k+1 \le 3$, so $k \le 2$. Choose a private vertex v of Q, let F be v with all off-Q branches rooted there, and let H be the complement. Then F is a tree and $Q - v$ is a path containing every shared cut of Q, so H is connected. Its only cycles are three triangles, including an intersecting pair, and therefore its cycle packing number is at most two. Equation (7) gives $\sigma(H) > 2$, whence

$$\sigma(G) \ge \sigma(F) + \sigma(H) > -1 + 2 = 1.$$

If no two designated triangles meet, connectedness of I forces every triangle to meet Q. No triangle can meet Q twice, since that would create a cycle in I, and no two can use the same cut. Thus they are exactly the three petals of Lemma 5.2. Each resulting triangular unicyclic territory has strictly positive surplus by (2); summing with (1) proves $\sigma(G) > 0$. □

6 Main theorem

Theorem 6.1. *Let G be a connected tetracyclic cactus on n vertices. Then*

$$\boxed{s^+(G) > n}.$$

Proof. The DNN estimate (8) settles every cycle multiset except (9). If the shared-cut graph is disconnected, Propositions 3.1 and 4.2 settle the two residual families. If it is connected, Propositions 5.3 and 5.1 settle them. These alternatives exhaust every cactus incidence. Every construction uses vertex-disjoint induced territories, so crossing edges are discarded only through (1); no edge monotonicity or tree relocation is used. □

7 Conclusion

Since a connected tetracyclic graph has $m = n + 3$, the strict theorem proves the conclusion of the Akbari–Kumar–Mohar–Pragada–Zhang conjecture [2, Conjecture 1.2] for every tetracyclic cactus, with arbitrary bridge blocks and arbitrary attached trees. A previously attempted coefficientwise phase certificate is unnecessary: the incidence-tree sacrifice proof above bypasses the fully shared core calculation entirely.

References

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[2] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.

[3] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, repository path `all-bicyclic-cacti/paper.tex`.

[4] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, repository path `all-tricyclic-cacti/paper.tex`.

[5] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, repository path `sharp-cactus-dnn/paper.tex`.

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[7] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.